Sensitivity Analyses for Unmeasured Confounding

Lucy D'Agostino McGowan

Wake Forest University

Recall: Propensity scores

Rosenbaum and Rubin showed in observational studies, conditioning on **propensity scores** can lead to unbiased estimates of the exposure effect

1. **There are no unmeasured confounders**
2. **Every subject has a nonzero probability of receiving either exposure**

How sensitive is our result to conditions other than those laid out in our assumptions and analysis?

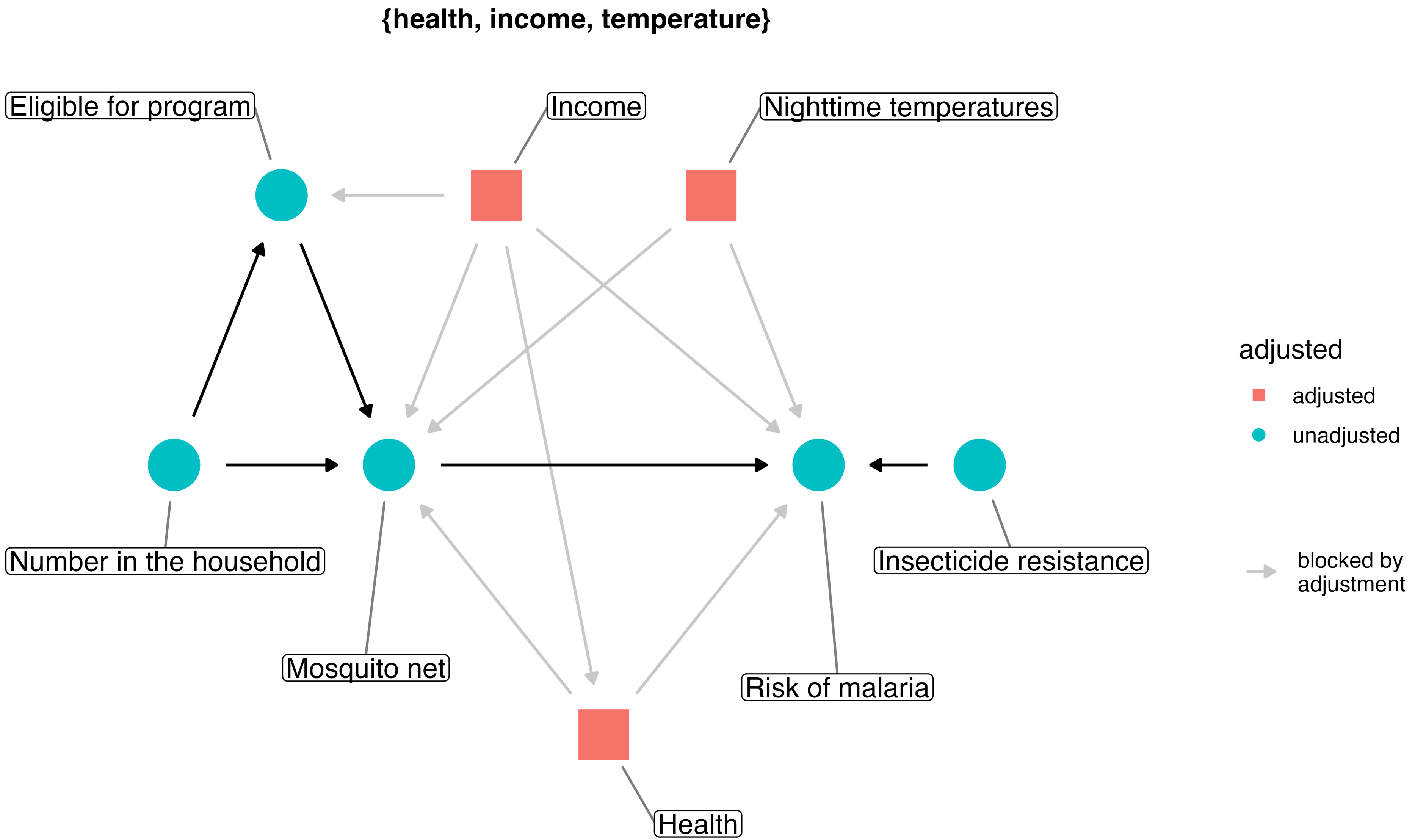
Two ways to probe a result

1. Explore the logical implications of the causal question and the DAG: does the data contradict what we assumed?

2. Use mathematical techniques to quantify how different the result would be under unmeasured confounding

Checking DAGs for robustness

Recall the mosquito net DAG



If the DAG is right, every valid set works

```r
library(dagitty)
adjustmentSets(mosquito_dag, type = "all")
```

If the DAG is right, every valid set works

```
{ health, income, temperature }
{ eligible, health, income, temperature }
{ health, household, income, temperature }
{ eligible, health, household, income, temperature }
{ health, income, resistance, temperature }
{ eligible, health, income, resistance, temperature }
{ health, household, income, resistance, temperature }
{ eligible, health, household, income, resistance, temperature }
```

Estimate the effect for one adjustment set

```
fit_ipw_effect <- function(.fmla, .data = net_data) {
  propensity_model <- glm(.fmla, data = .data, family = binomial())

  propensity_model |>
    augment(type.predict = "response", data = .data) |>
    mutate(wts = wt_ate(.fitted, net, exposure_type = "binary")) |>
    lm(malaria_risk ~ net, data = _, weights = wts) |>
    tidy() |>
    filter(term == "netTRUE") |>
    pull(estimate)
}
```

Estimate the effect for four of them

```r
adjustment_sets <- list(
  "income, health, temperature" = net ~ income + health + temperature,
  "+ eligible" = net ~ income + health + temperature + eligible,
  "+ household" = net ~ income + health + temperature + household,
  "+ resistance" = net ~ income + health + temperature +
    insecticide_resistance
)

tibble(
  adjustment_set = names(adjustment_sets),
  ate = map_dbl(adjustment_sets, fit_ipw_effect)
)
```

Estimate the effect for four of them

```
# A tibble: 4 × 2
  adjustment_set                  ate
  <chr>                         <dbl>
1 income, health, temperature -12.5
2 + eligible                    -12.9
3 + household                   -12.7
4 + resistance                  -12.7
```

Now suppose income had not been measured

```r
fit_ipw_effect(net ~ health + temperature)
```

```
[1] -14.23818
```

Leaving out a confounder moves the estimate well outside the spread of the valid sets

Agreement is evidence, not proof

A DAG can be right about every variable it contains and still be missing one entirely

Negative controls

Negative controls

An exposure or outcome similar to your question in as many ways as possible, except that there **should not** be a causal effect

1. Leave out an essential ingredient
2. Inactivate the hypothesized active ingredient
3. Check for an effect impossible by the hypothesized mechanism

A lagged negative exposure

`net_data` is a single cross-section, so we'll use the Seven Dwarfs data, which is also the data for your turn

If Extra Magic Mornings raise wait times, that effect should not still be there nine weeks later

Lag the exposure and its confounders

```r
lag_seven_dwarfs <- function(n_days_lag) {
  seven_dwarfs |>
    arrange(park_date) |>
    transmute(
      park_date,
      prev_emm = lag(park_extra_magic_morning, n_days_lag),
      prev_temperature = lag(park_temperature_high, n_days_lag),
      prev_close = lag(park_close, n_days_lag),
      prev_season = lag(park_ticket_season, n_days_lag)
    ) |>
    left_join(seven_dwarfs, by = "park_date") |>
    drop_na(prev_emm)
}
```

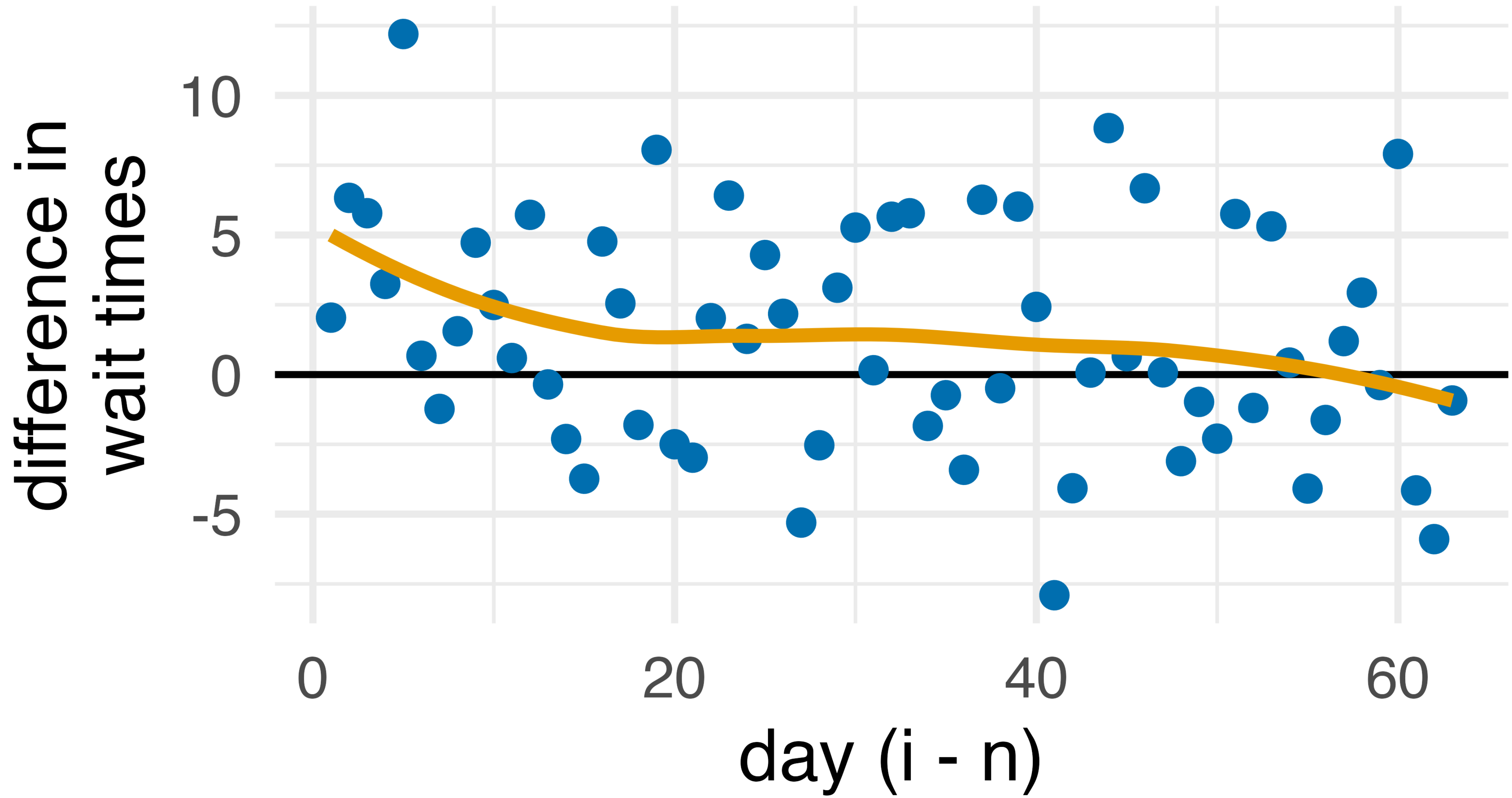
Fit the effect 63 days back

```r
fit_lagged_effect <- function(n_days_lag) {
  lagged <- lag_seven_dwarfs(n_days_lag)

  propensity_model <- glm(
    prev_emm ~ prev_temperature + prev_close + prev_season,
    data = lagged,
    family = binomial()
  )

  propensity_model |>
    augment(type.predict = "response", data = lagged) |>
    mutate(wts = wt_ate(.fitted, prev_emm, exposure_type = "binary")) |>
    lm(
      # day i's exposure inactivates the indirect effect
      wait_minutes_posted_avg ~ prev_emm + park_extra_magic_morning,
      data = _,
      weights = wts
    ) |>
    tidy() |>
    filter(term == "prev_emm") |>
    pull(estimate)
}

fit_lagged_effect(63)
```

Fit the effect 63 days back

```
[1] -0.9285462
```

The effect should decay to null



A lingering effect

In the outcome model exercises, the effect on day i was about 6 minutes

The lagged effect eventually approaches null, but it hovers above null for far longer than any real effect of Extra Magic Mornings could, which implies residual confounding

Negative outcomes

- Wait times at Universal Studios share Orlando's weather and crowds, but Extra Magic Mornings cannot cause them
- Simulated Universal wait times with no shared confounder give a null effect
- Adding a confounder that drives wait times at both parks makes them non-null too

DAG-data consistency

What does the DAG say should be null?

```r
query_conditional_independence(mosquito_dag) |>
  unnest(conditioned_on)
```

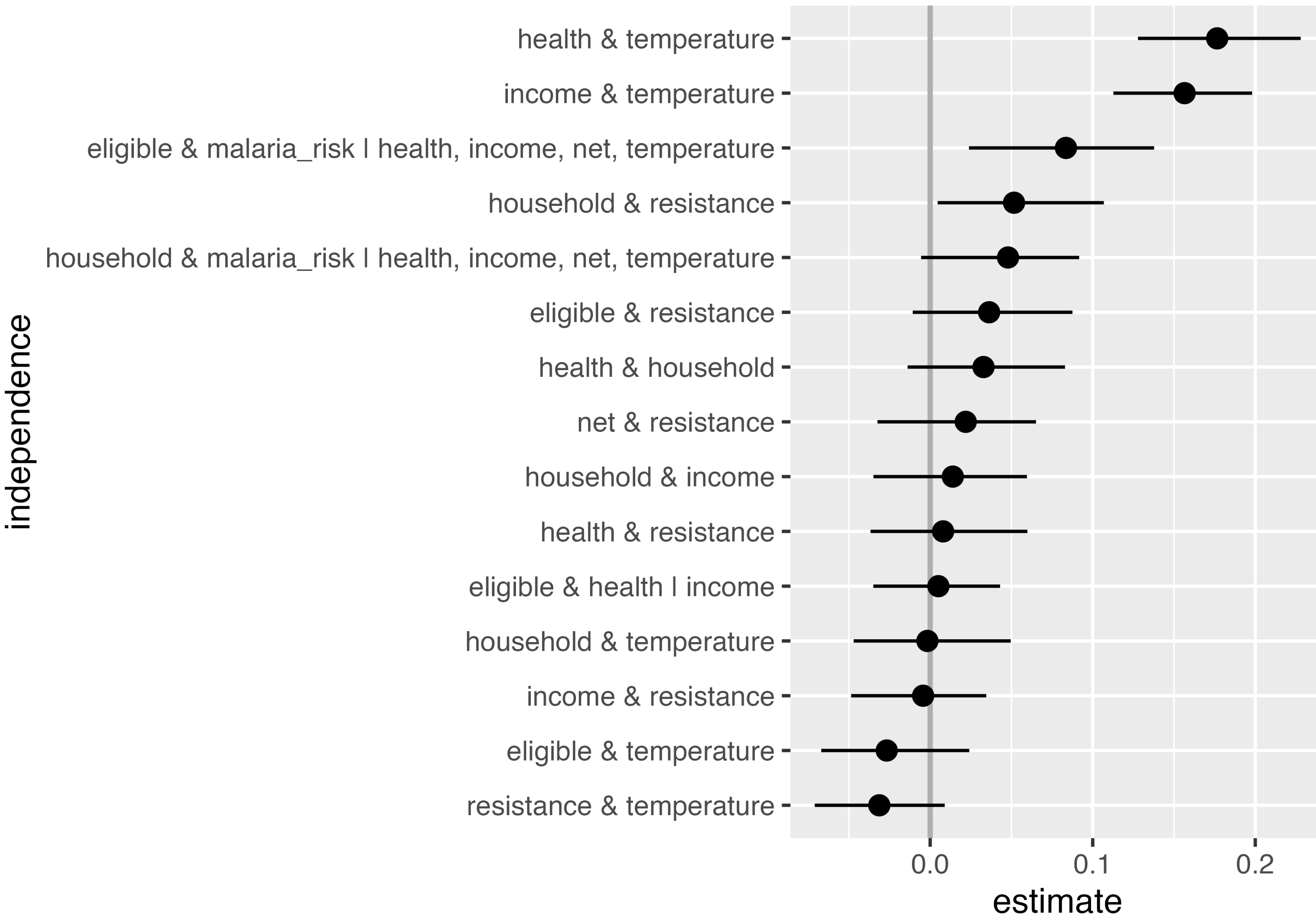
What does the DAG say should be null?

```
# A tibble: 21 × 5
   set   a        b           conditioning_set conditioned_on
   <chr> <chr>    <chr>       <chr>            <chr>
 1 1     eligible health      {income}         income
 2 2     eligible malaria_r… {health, income… health
 3 2     eligible malaria_r… {health, income… income
 4 2     eligible malaria_r… {health, income… net
 5 2     eligible malaria_r… {health, income… temperature
 6 3     eligible resistance <NA>             <NA>
 7 4     eligible temperatu… <NA>             <NA>
 8 5     health   household  <NA>             <NA>
 9 6     health   resistance <NA>             <NA>
10 7     health   temperatu… <NA>             <NA>
# ℹ 11 more rows
```

Do those relationships hold in the data?

```r
test_conditional_independence(
  mosquito_dag,
  data = net_data |>
    transmute(
      net = as.numeric(net),
      malaria_risk,
      income,
      health,
      temperature,
      resistance = insecticide_resistance,
      eligible = as.numeric(eligible),
      household
    ) |>
    as.data.frame(),
  # bootstrapped samples for the confidence intervals
  R = 200
) |>
  ggdag_conditional_independence()
```

Do those relationships hold in the data?



Four of them do not hold

- The DAG says temperature is independent of income and of health; in the data they correlate at 0.16 and 0.18

- Two weaker failures: `eligible` and malaria risk, at 0.08 even after conditioning on the adjustment set, and `household` and insecticide resistance, at 0.05

- Adding arrows for the temperature pairs leaves the adjustment set unchanged, but report the original DAG if you change it

These tests cannot prove a DAG right or wrong; they tell us where the data disagrees with our assumptions

Quantifying Unmeasured Confounding

Current Epidemiology Reports (2022) 9:361–375
https://doi.org/10.1007/s40471-022-00308-6

EPIDEMIOLOGIC METHODS (P P HOWARDS, SECTION EDITOR)

Sensitivity Analyses for Unmeasured Confounders

Lucy D'Agostino McGowan[1]

Accepted: 15 September 2022 / Published online: 8 November 2022

The Journal of Open Source Software

tipr: An R package for sensitivity analyses for unmeasured confounders

Lucy D'Agostino McGowan [1¶]

1 Wake Forest University, USA ¶ Corresponding author

DOI: 10.21105/joss.04495

Quantifying Unmeasured Confounding

1. The exposure-outcome effect
2. The exposure-unmeasured confounder effect
3. The unmeasured confounder-outcome effect

Quantifying Unmeasured Confounding

Table 4 Sensitivity analysis equations and R code

Outcome	Effect of Interest	Sensitivity analysis equation	R function from tipr package
Unmeasured confounder, U, is Normally distributed with a difference in means between exposure groups of d, unit variance, and association with Y of $\beta_{Y\sim U\mid X+\mathbf{Z}}$			
Continuous	Coefficient	$\beta_{Y\sim X\mid U+\mathbf{Z}} = \beta_{Y\sim X\mid\mathbf{Z}} - \beta_{Y\sim U\mid X+\mathbf{Z}} \times d$	`adjust_coef()`
Binary[a]	Coefficient	$\beta_{Y\sim X\mid U+\mathbf{Z}} = \beta_{Y\sim X\mid\mathbf{Z}} - \beta_{Y\sim U\mid X+\mathbf{Z}} \times d$	`adjust_coef()`
Time to event	Coefficient	$\beta_{Y\sim X\mid U+\mathbf{Z}} \approx \beta_{Y\sim X\mid\mathbf{Z}} - \beta_{Y\sim U\mid X+\mathbf{Z}} \times d$	`adjust_coef()`
Binary	Risk ratio	$RR_{Y\sim X\mid U+\mathbf{Z}} = \frac{RR_{Y\sim X\mid\mathbf{Z}}}{RR^{d}_{Y\sim U\mid X+\mathbf{Z}}}$	`adjust_rr()`
Binary	Odds ratio	$OR_{Y\sim X\mid U+\mathbf{Z}} \approx \frac{OR_{Y\sim X\mid\mathbf{Z}}}{OR^{d}_{Y\sim U\mid X+\mathbf{Z}}}$	`adjust_or()`
Time to event	Hazard ratio	$HR_{Y\sim X\mid U+\mathbf{Z}} \approx \frac{HR_{Y\sim X\mid\mathbf{Z}}}{HR^{d}_{Y\sim U\mid X+\mathbf{Z}}}$	`adjust_hr()`
Unmeasured confounder, U, is binary with prevalence in the unexposed group of p_0 in the exposed group of p_1, and association with Y of $\beta_{Y\sim U\mid X+\mathbf{Z}}$			
Continuous	Coefficient	$\beta_{Y\sim X\mid U+\mathbf{Z}} = \beta_{Y\sim X\mid\mathbf{Z}} - \beta_{Y\sim U\mid X+\mathbf{Z}} \times (p_1 - p_0)$	`adjust_coef_with_binary()`
Binary[a]	Coefficient	$\beta_{Y\sim X\mid U+\mathbf{Z}} = \beta_{Y\sim X\mid\mathbf{Z}} - log\frac{e^{\beta_{Y\sim U\mid X+\mathbf{Z}}}p_1+(1-p_1)}{e^{\beta_{Y\sim U\mid X+\mathbf{Z}}}p_0+(1-p_0)}$	`adjust_coef_with_binary()`
Time to event	Coefficient	$\beta_{Y\sim X\mid U+\mathbf{Z}} \approx \beta_{Y\sim X\mid\mathbf{Z}} - log\frac{e^{\beta_{Y\sim U\mid X+\mathbf{Z}}}p_1+(1-p_1)}{e^{\beta_{Y\sim U\mid X+\mathbf{Z}}}p_0+(1-p_0)}$	`adjust_coef_with_binary()`
Binary	Risk ratio	$RR_{Y\sim X\mid U+\mathbf{Z}} = RR_{Y\sim X\mid\mathbf{Z}}\frac{RR_{Y\sim U\mid X+\mathbf{Z}}p_0+(1-p_0)}{RR_{Y\sim U\mid X+\mathbf{Z}}p_1+(1-p_1)}$	`adjust_rr_with_binary()`
Binary	Odds ratio	$OR_{Y\sim X\mid U+\mathbf{Z}} \approx OR_{Y\sim X\mid\mathbf{Z}}\frac{OR_{Y\sim U\mid X+\mathbf{Z}}p_0+(1-p_0)}{OR_{Y\sim U\mid X+\mathbf{Z}}p_1+(1-p_1)}$	`adjust_or_with_binary()`
Time to event	Hazard ratio	$HR_{Y\sim X\mid U+\mathbf{Z}} \approx HR_{Y\sim X\mid\mathbf{Z}}\frac{HR_{Y\sim U\mid X+\mathbf{Z}}p_0+(1-p_0)}{HR_{Y\sim U\mid X+\mathbf{Z}}p_1+(1-p_1)}$	`adjust_hr_with_binary()`
Unmeasured confounder, U, has relationships with X and Y characterized by $\beta_{U\sim X\mid\mathbf{Z}}$ and $\beta_{Y\sim U\mid X+\mathbf{Z}}$ or by partial R^2 as $R^2_{X\sim U\mid\mathbf{Z}}$ and $R^2_{Y\sim U\mid X+\mathbf{Z}}$			
Continuous	Coefficient	$\beta_{Y\sim X\mid U+\mathbf{Z}} = \beta_{Y\sim X\mid\mathbf{Z}} - \beta_{Y\sim U\mid X+\mathbf{Z}} \times \beta_{U\sim X\mid\mathbf{Z}}$	`adjust_coef()`
Continuous	Coefficient	$\beta_{Y\sim X\mid U+\mathbf{Z}} = \beta_{Y\sim X\mid\mathbf{Z}} - \mathrm{se}(\beta_{Y\sim X\mid\mathbf{Z}})\sqrt{\frac{R^2_{Y\sim U\mid X+\mathbf{Z}}\times R^2_{X\sim U\mid\mathbf{Z}}}{(1-R^2_{X\sim U\mid\mathbf{Z}})}\mathrm{df}}$	`adjust_coef_with_r2()`

[a] The equality will hold for binary outcomes fit via a loglinear model. Otherwise, this is an approximation

D'Agostino McGowan, L. Sensitivity Analyses for Unmeasured Confounders. Curr Epidemiol Rep 9, 361–375 (2022)

Quantifying Unmeasured Confounding

Table 7 Tipping point equations and R code

From: Sensitivity Analyses for Unmeasured Confounders

Effect of Interest	Known relationship	Tipping point equation	R function from the `tipr` package
Unmeasured confounder, *U*, is Normally distributed with a difference in means between exposure groups of *d*, unit variance, and association with *Y* of $\beta_{Y\sim U\|X+\mathbf{Z}}$			
Coefficient	*U* and *X*	$\beta_{Y\sim U\|X+\mathbf{Z},tip=0} = \frac{\beta_{Y\sim X\|\mathbf{Z}}}{d}$	tip_coef()
Coefficient	*U* and *Y*	$d_{tip=0} = \frac{\beta_{Y\sim X\|\mathbf{Z}}}{\beta_{Y\sim U\|X+\mathbf{Z}}}$	tip_coef()
Risk ratio	*U* and *X*	$RR_{Y\sim U\|X+\mathbf{Z},tip=1} = RR^{1/d}_{Y\sim X\|\mathbf{Z}}$	tip_rr() tip_or() tip_hr()
Risk ratio	*U* and *Y*	$d_{tip=1} = \frac{\log(RR_{Y\sim X\|\mathbf{Z}})}{\log(RR_{Y\sim U\|X+\mathbf{Z}})}$	tip_rr() tip_or() tip_hr()
Unmeasured confounder, *U*, is binary with prevalence in the unexposed group of p_0 in the exposed group of p_1, and association with *Y* of $\beta_{Y\sim U\|X+\mathbf{Z}}$			
Risk ratio	*U* and *X*	$RR_{Y\sim U\|X+\mathbf{Z},tip=1} = \frac{(1-p_1)+RR_{Y\sim X\|\mathbf{Z}}\times(p_0-1)}{RR_{Y\sim X\|\mathbf{Z}}\times p_0-p_1}$	tip_rr_with_binary() tip_or_with_binary() tip_hr_with_binary()
Risk ratio	*U* and *Y*	$p_{1,tip=1} = \frac{RR_{Y\sim X\|\mathbf{Z}}-1}{RR_{Y\sim U\|X+\mathbf{Z}}-1} + (RR_{Y\sim X\|\mathbf{Z}} \times p_0)$ $p_{0,tip=1} = \frac{p_1}{RR_{Y\sim X\|\mathbf{Z}}} - \frac{RR_{Y\sim X\|\mathbf{Z}}-1}{RR_{Y\sim X\|\mathbf{Z}}\times(RR_{Y\sim U\|X+\mathbf{Z}}-1)}$	tip_rr_with_binary() tip_or_with_binary() tip_hr_with_binary()
Unmeasured confounder, *U*, has relationships with *X* and *Y* characterized by $\beta_{U\sim X\|\mathbf{Z}}$ and $\beta_{Y\sim U\|X+\mathbf{Z}}$ or by partial R^2 as $R^2_{X\sim U\|\mathbf{Z}}$ and $R^2_{Y\sim U\|X+\mathbf{Z}}$			
Coefficient	*U* and *X*	$\beta_{Y\sim U\|X+\mathbf{Z},tip=0} = \frac{\beta_{Y\sim X\|\mathbf{Z}}}{\beta_{U\sim X\|\mathbf{Z}}}$	tip_coef()
Coefficient	*U* and *Y*	$\beta_{U\sim X\|\mathbf{Z},tip=0} = \frac{\beta_{Y\sim X\|\mathbf{Z}}}{\beta_{Y\sim U\|X+\mathbf{Z}}}$	tip_coef()
Coefficient	*U* and *X*	$R^2_{Y\sim U\|X+\mathbf{Z},tip=0} = \frac{\beta^2_{Y\sim X\|\mathbf{Z}}-\beta^2_{Y\sim X\|\mathbf{Z}}\times R^2_{X\sim U\|\mathbf{Z}}}{se^2(\beta_{Y\sim X\|\mathbf{Z}})df\times R^2_{X\sim U\|\mathbf{Z}}}$	tip_coef _with_r2()
Coefficient	*U* and *Y*	$R^2_{X\sim U\|\mathbf{Z},tip=0} = \frac{\beta^2_{Y\sim X\|\mathbf{Z}}}{\beta^2_{Y\sim X\|\mathbf{Z}}+se^2(\beta_{Y\sim X\|\mathbf{Z}})df\times R^2_{Y\sim U\|X+\mathbf{Z}}}$	tip_coef _with_r2()

D'Agostino McGowan, L. Sensitivity Analyses for Unmeasured Confounders. Curr Epidemiol Rep 9, 361–375 (2022)

tipr

The grammar of tipr: `{action}_{effect}_with_{what}`

part	term	what it means
action	`adjust`	needs both confounder relationships
	`tip`	needs only one of them
effect	`coef`, `rr`, `or`, `hr`	the scale of `effect_observed`
what	`continuous`	`exposure_confounder_effect`, `confounder_outcome_effect`
	`binary`	`exposed_confounder_prev`, `unexposed_confounder_prev`, `confounder_outcome_effect`
	`r2`	`confounder_exposure_r2`, `confounder_outcome_r2`

D'Agostino McGowan, L., (2022). tipr: An R package for sensitivity analyses for unmeasured confounders. Journal of Open Source Software, 7(77), 4495

What is known about the unmeasured confounder?

- **Both** the exposure and outcome relationships: `adjust_*()` functions

- **One** of the two: `tip_*()` functions, or `adjust_*()` functions in an array

- **Neither**: `adjust_*()` or `tip_*()` in an array, `tip_coef_with_r2()` benchmarked against measured confounders, or summaries like `r_value()` and `e_value()`

1. The observed exposure-outcome effect

```r
propensity_model <- glm(
  net ~ income + health + temperature,
  family = binomial(),
  data = net_data
)

net_data_wts <- propensity_model |>
  augment(type.predict = "response", data = net_data) |>
  mutate(wts = wt_ate(.fitted, net, exposure_type = "binary"))

observed_effect <- ipw(
  propensity_model,
  lm(malaria_risk ~ net, data = net_data_wts, weights = wts)
) |>
  tidy(conf.int = TRUE)

observed_effect |>
  select(estimate, conf.low, conf.high)
```

1. The observed exposure-outcome effect

```
# A tibble: 1 × 3
  estimate conf.low conf.high
     <dbl>    <dbl>     <dbl>
1    -12.5    -13.4     -11.7
```

2. The confounder-exposure relationship

- **Binary** confounder: its prevalence in the exposed and in the unexposed

- **Continuous** confounder: the difference in means, assuming a normal distribution and unit variance

- **Distribution-agnostic**: the partial R^2 for the exposure

What “unit variance” means

```r
temperature_sd <- sd(net_data$temperature)

net_data |>
  group_by(net) |>
  summarize(
    mean = mean(temperature),
    # divide by the SD to put the confounder on a unit-variance scale
    standardized_mean = mean / temperature_sd
  )
```

What “unit variance” means

```
# A tibble: 2 × 3
  net    mean standardized_mean
  <lgl> <dbl>             <dbl>
1 FALSE  24.0              5.86
2 TRUE   23.7              5.79
```

What “unit variance” means

- Nighttime temperatures average 23.7 degrees for net users and 24.0 for everyone else, with a standard deviation of 4.1 degrees

- On a unit-variance scale that difference is -0.08

- So `exposure_confounder_effect = 0.1` means the confounder differs by a tenth of a standard deviation between exposure groups

3. The confounder-outcome relationship

- The coefficient for the confounder in a fully adjusted outcome model, in the units of the outcome
- Here that is points of malaria risk per standard deviation of the confounder
- Or the exponentiated version: a risk ratio, odds ratio, or hazard ratio

An unmeasured confounder: genetic resistance

1. People with this genetic resistance have a malaria risk lower by about 10
2. About 26% of people who use nets have this genetic resistance
3. About 5% of people who don't use nets have this genetic resistance

Adjust the estimate for genetic resistance

```r
observed_effect |>
  select(estimate, conf.low, conf.high) |>
  unlist() |>
  adjust_coef_with_binary(
    exposed_confounder_prev = 0.26,
    unexposed_confounder_prev = 0.05,
    confounder_outcome_effect = -10
  ) |>
  select(effect_observed, effect_adjusted)
```

Adjust the estimate for genetic resistance

```
# A tibble: 3 × 2
  effect_observed effect_adjusted
            <dbl>           <dbl>
1           -12.5          -10.4 
2           -13.4          -11.3 
3           -11.7           -9.59
```

These data are simulated. The true effect of net use on malaria risk is about -10

Bias analysis, checked against a known answer

- Observed, with genetic resistance unmeasured: **-12.5 (-13.4, -11.7)**
- Adjusted with tipr: **-10.4 (-11.3, -9.6)**
- Re-estimated with genetic resistance measured: **-10.2**
- The truth: **-10**

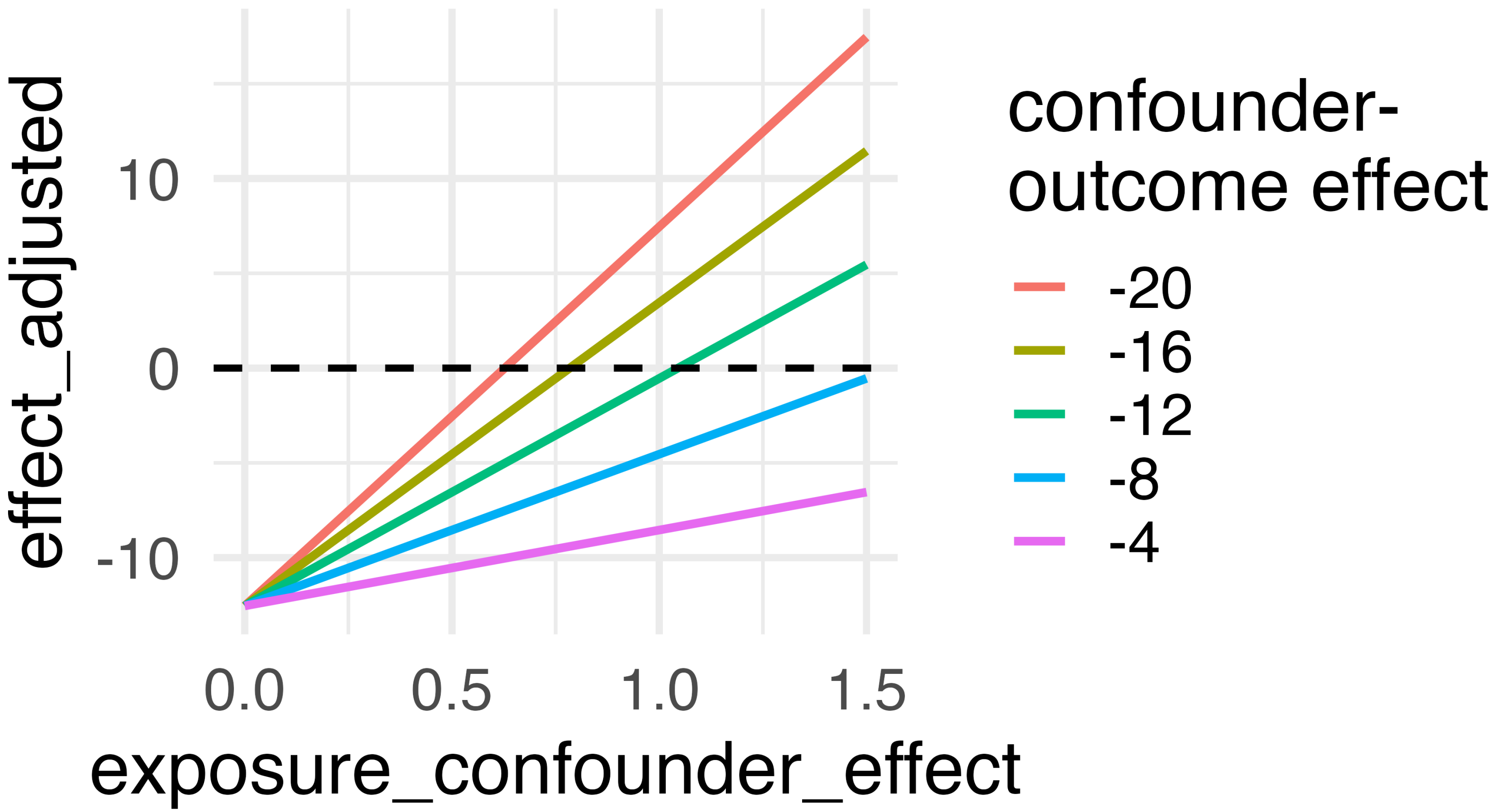
Vary both parameters at once

```r
sens <- adjust_coef(
  effect_observed = observed_effect$estimate,
  exposure_confounder_effect = rep(seq(0, 1.5, by = 0.05), each = 5),
  confounder_outcome_effect = rep(seq(-4, -20, by = -4), times = 31),
  verbose = FALSE
)
```

Vary both parameters at once

```r
ggplot(
  sens,
  aes(
    exposure_confounder_effect,
    effect_adjusted,
    color = factor(confounder_outcome_effect)
  )
) +
  geom_line(linewidth = 1.1) +
  geom_hline(yintercept = 0, lty = 2) +
  theme_minimal(base_size = 20) +
  labs(color = "confounder-\noutcome effect")
```

Vary both parameters at once



What will tip our confidence bound to cross zero?

`tip_coef()` solves for the crossing

```r
tip_coef(
  effect_observed = observed_effect$conf.high,
  confounder_outcome_effect = -10
) |>
  select(exposure_confounder_effect)
```

`tip_coef()` solves for the crossing

```
# A tibble: 1 × 1
  exposure_confounder_effect
                       <dbl>
1                       1.17
```

“An unmeasured confounder with an effect of *-10* on malaria risk would have to differ by *1.17* standard deviations between net users and non-users to tip our upper bound to zero. Genetic resistance differs in prevalence by about *0.21*, so a confounder like it cannot on its own explain away this result.”

The same grammar, in the wild

Cancer Causes & Control (2018) 29:823–832
https://doi.org/10.1007/s10552-018-1058-4

ORIGINAL PAPER

Metformin use and incidence cancer risk: evidence for a selective protective effect against liver cancer

Harvey J. Murff[1,2,5] · Christianne L. Roumie[1,2] · Robert A. Greevy[1,3] · Amber J. Hackstadt[3] · Lucy E. D'Agostino McGowan[3] · Adriana M. Hung[1,2] · Carlos G. Grijalva[1,4] · Marie R. Griffin[1,2,4]

Received: 22 April 2017 / Accepted: 13 July 2018 / Published online: 18 July 2018

The published result

- **Matched** 42,217 new metformin users to 42,217 new sulfonylurea users
- **Outcome:** Lung Cancer
- **Adjusted Hazard Ratio**: 0.87 (0.79, 0.96)

What if heavy alcohol consumption is prevalent among 4% of Metformin users and 6% of Sulfonylurea users?

Meadows SO, Engel CC, Collins RL, Beckman RL, Cefalu M, Hawes-Dawson J, et al. 2015 health related behaviors survey: Substance use among US active-duty service members. RAND; 2018.

tipr Example

What if we assume the effect of alcohol consumption on lung cancer after adjusting for other confounders is 2?

```
adjust_hr_with_binary(
  effect_observed = c(0.79, 0.87, 0.96),
  exposed_confounder_prev = 0.04,
  unexposed_confounder_prev = 0.06,
  confounder_outcome_effect = 2
)
```

tipr Example

```
# A tibble: 3 × 5
  hr_adjusted hr_observed exposed_confounder_prev
        <dbl>       <dbl>                   <dbl>
1       0.805        0.79                    0.04
2       0.887        0.87                    0.04
3       0.978        0.96                    0.04
# ℹ 2 more variables: unexposed_confounder_prev <dbl>,
#   confounder_outcome_effect <dbl>
```

tipr Example

```
tip_hr_with_binary(
  effect_observed = 0.96,
  exposed_confounder_prev = 0.04,
  unexposed_confounder_prev = 0.06
) |>
  select(confounder_outcome_effect)
```

`tipr` Example

```
# A tibble: 1 × 1
  confounder_outcome_effect
                      <dbl>
1                      3.27
```

“If heavy alcohol consumption differed between groups, with *4%* prevalence among metformin users and *6%* among sulfonylureas users, it would need to have an association with lung cancer incidence of *3.27 to tip this analysis* at the 5% level, rendering it inconclusive. This effect is larger than the understood association between lung cancer and alcohol consumption.”

When nothing is known: `e_value()`

```r
# the smallest confounder-exposure and confounder-outcome
# association that could explain away the observed effect
e_value(0.96)
```

```
[1] 1.25
```

Your turn

Use the `tip_coef()` function to conduct a sensitivity analysis for the estimate you calculated in `exercises/10-outcome-model-exercises.qmd`. Use the lower bound of the confidence interval for the effect and `0.1` for the exposure-confounder effect.

05:00